

Credit Risk Sensitivity Analysis

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1 Introduction

Retail lenders collect a broad set of variables to support credit default models: credit limits, monthly billing statements, payment amounts, repayment status codes, and demographic characteristics. Assembling and maintaining these data sources is operationally expensive, and the marginal value of each is rarely tested rigorously. A billing statement from six months ago may carry information already captured by last month’s repayment status. When correlated data sources are included without scrutiny, the result is coefficient instability, unnecessary model complexity, and data infrastructure costs that may not be justified by predictive improvement.

The goal is to investigate which data sources carry independent predictive signal for credit default given what other variables already capture, with direct implications for data storage efficiency and pipeline design. We compare nested logistic regression specifications across variable groups, use LASSO and Ridge regularization paths to rank the independent contribution of each data source, test whether that contribution varies across delinquency regimes using interaction models, and benchmark the most parsimonious specification against XGBoost to quantify the cost of simplification.

2 Data and Domain Problem

Data is sourced from UC Irvine’s Default of Credit Card Clients dataset, collected from a major Taiwanese commercial bank, covering April to September 2005 with default in October 2005 as the binary outcome. The overall default rate is 22.1%, motivating AUC over accuracy as the primary evaluation metric. Predictors fall into financial variables (credit limit, bill amounts, payment amounts), behavioral variables (six monthly repayment status codes), and demographics. Each group has different data collection costs, and whether each justifies those costs is the central question.

The PAY block has within-block correlations exceeding 0.7, so we aggregate delinquency into three summaries: occurrence (ANY_DELAY), severity (MAX_DELAY), and frequency (NUM_DELAY_MONTHS). We also test whether each data source’s contribution varies across delinquency segments via a delinquency indicator and full interaction terms, compared against a pooled baseline with a likelihood ratio test. EDA confirmed that default rates rise sharply with delinquency severity (13% to over 75%), financial variables do not update after account opening, and the bill amount block has VIFs up to 25.6. Full figures are in Appendix 7.

3 Models and Results

We compare logistic regression specifications across nested variable groups on a stratified 20% held-out test set with 5-fold cross-validation. We also evaluate regularized and interaction-based models. Table 1 summarizes all results.

Table 1: Model performance across specifications and methods.

Model	Type	CV AUC	Test AUC
Financial + # delayed months	Logistic	0.751	0.735
# delayed months + demographics	Logistic	0.739	0.728
Financial + max delay	Logistic	0.732	0.716
Behavioral (# delayed months)	Logistic	0.726	0.713
Full with demographics	Logistic	0.724	0.710
Financial + behavioral (full PAY)	Logistic	0.722	0.706
Behavioral (max delay)	Logistic	0.708	0.691
Behavioral (full PAY)	Logistic	0.705	0.690
Behavioral (any delay)	Logistic	0.701	0.688
Financial only	Logistic	0.664	0.649
LASSO (CV-optimal)	Regularized	–	0.740
Ridge (CV-optimal)	Regularized	–	0.740
Restricted (DELINQ indicator)	Logistic	–	0.737
Unrestricted (full interactions)	Logistic	–	0.738
XGBoost (full features)	Nonlinear	0.783	0.775

Repayment behavior outperforms financial variables in isolation (CV AUC 0.705 vs. 0.664), and granular repayment codes add nothing beyond a scalar count: replacing six monthly status codes with NUM_DELAY_MONTHS improves performance from 0.705 to 0.726, while the full 22-variable combined model achieves only 0.722. The strongest model combines NUM_DELAY_MONTHS with financial variables to reach CV AUC 0.751 with 14 inputs, suggesting six months of granular repayment codes can be dropped without performance loss.

The regularization paths address the billing data source directly. Delinquency variables enter the LASSO path earliest with the largest coefficients; bill amount variables enter last with near-zero Ridge coefficients throughout, consistent with pairwise correlations up to 0.95. Both regularized models achieve test AUC of 0.740. Full plots are in Appendix 9.

To test whether data source value varies by segment, we add a delinquency indicator and full interaction terms, comparing the extended model against a restricted baseline via likelihood ratio test. The test rejects the restricted model ($p < 0.001$). The predictive improvement is modest (test AUC 0.737 and 0.738), but interaction coefficients show billing balance variables become meaningful among delinquent borrowers. Full regime analysis is in Appendix 11.

XGBoost achieves test AUC of 0.775, a gap of 0.040 above the best logistic specification. Feature importances corroborate regularization: behavioral summaries dominate at 0.572 total importance while bill amounts contribute only 0.061. Full figures are in Appendix 12.

4 Discussion of Results

Not all data sources are mission-critical. Behavioral covariates, specifically delinquency frequency, dominate default prediction. Granular repayment codes are redundant once that count is included, and the billing statement block contributes negligible independent signal in the pooled model. Credit limit and delinquency count achieve essentially the same discrimination as the full feature set.

The regime interaction model adds an important qualification. Billing balances are redundant for current accounts but become meaningfully predictive among delinquent borrowers, where recent balances signal charge-off risk. The billing data source can be deprioritized for current-account screening but warrants retention for collections triage. The sharpest early warning signal occurs at the transition from zero to one month past due, where default rates jump from 13% to 34%.

When false negatives and false positives carry different costs, 0.5 is not the right threshold. At a cost ratio of 5:1, the appropriate threshold drops to 0.177, recovering 62.5% recall. Operating at 0.5 would miss a substantial share of true defaulters.

One limitation is temporal scope. Six months of data may capture seasonality or a specific credit cycle phase rather than long-run relationships, and data source importance could shift over a full economic cycle.

5 Conclusion

Credit default models can be substantially simplified without meaningful loss of predictive power. A model built on credit limit and a count of missed payments comes within 0.040 AUC of XGBoost on the full feature set, and outperforms models including six months of granular repayment codes and billing history. The data sources lenders invest most heavily in assembling contribute the least independent signal once simpler summaries are included. The regime interaction model adds nuance rather than contradiction: billing balances matter for delinquent borrowers, suggesting data strategy should vary by credit lifecycle stage rather than applying a uniform feature set across all accounts.

Additional Work

6 Model Selection and Assumptions

We use logistic regression as the primary framework: the binary outcome and fixed one-month prediction horizon make it a natural fit, and its probability estimates are directly interpretable as default risk scores. A probit link would yield nearly identical results. We use stratified 5-fold cross-validation with all preprocessing fit exclusively on training folds to prevent leakage. LASSO and Ridge are presented separately to isolate variable selection from coefficient stabilization. XGBoost is included with fixed hyperparameters as a nonlinear performance ceiling.

7 Exploratory Data Analysis

7.1 Behavioral Variables and Default

Figure 1 shows default rates by repayment status and delinquency count. The default rate is 13% for on-time borrowers, rising to 34% at one month late, 69% at two months, and exceeding 75% at three or more. The `NUM_DELAY_MONTHS` relationship is monotonically increasing from 12% to 70%, confirming that the scalar count captures the primary predictive content of the `PAY` block. The sharp jump at one month past due suggests that early intervention at this transition could meaningfully reduce portfolio-level losses.

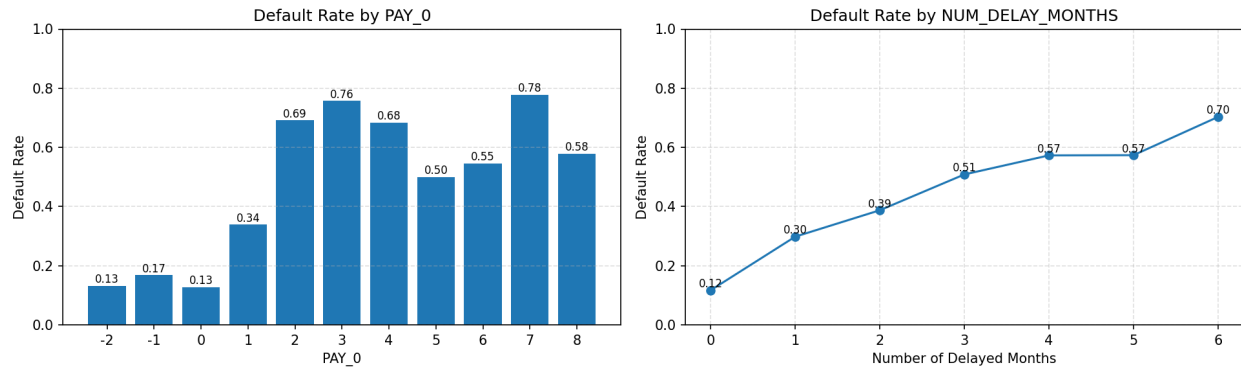


Figure 1: Default rate by `PAY_0` (left) and `NUM_DELAY_MONTHS` (right).

7.2 Financial Variables and Default

Figure 2 shows default rates fall from 36% to 12% across credit limit deciles, but the gradient is substantially shallower than for behavioral variables. Credit limits reflect capacity assessments at origination rather than current repayment distress, making them less discriminating for near-term default prediction.

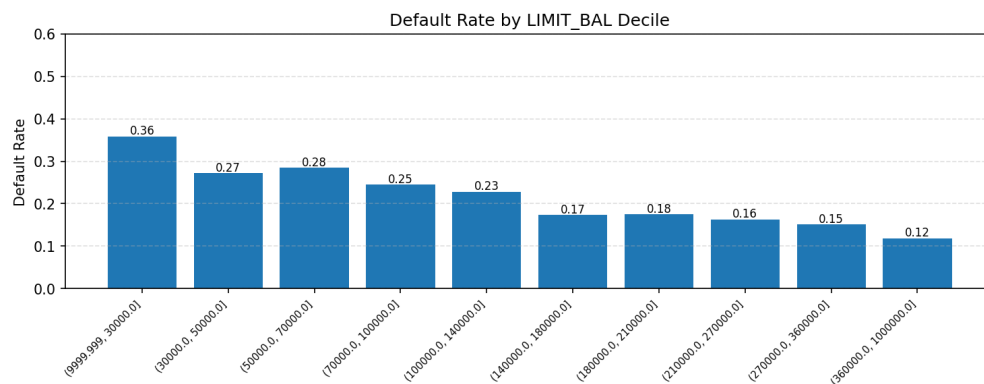


Figure 2: Default rate by credit limit decile.

7.3 Multicollinearity Structure

Figure 3 reports VIFs for selected predictors. The `BILL_AMT` block has pairwise correlations up to 0.95 and VIFs exceeding 25. `LIMIT_BAL` ($VIF = 1.4$) and `NUM_DELAY_MONTHS` ($VIF = 2.5$) are

near-orthogonal to other predictors, explaining their dominance in the LASSO path.

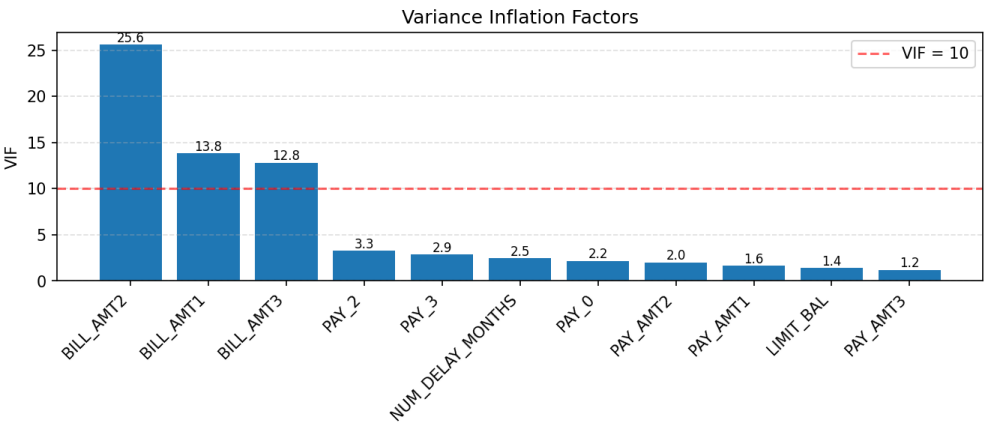


Figure 3: Variance inflation factors for selected predictors.

8 Model Performance Figures

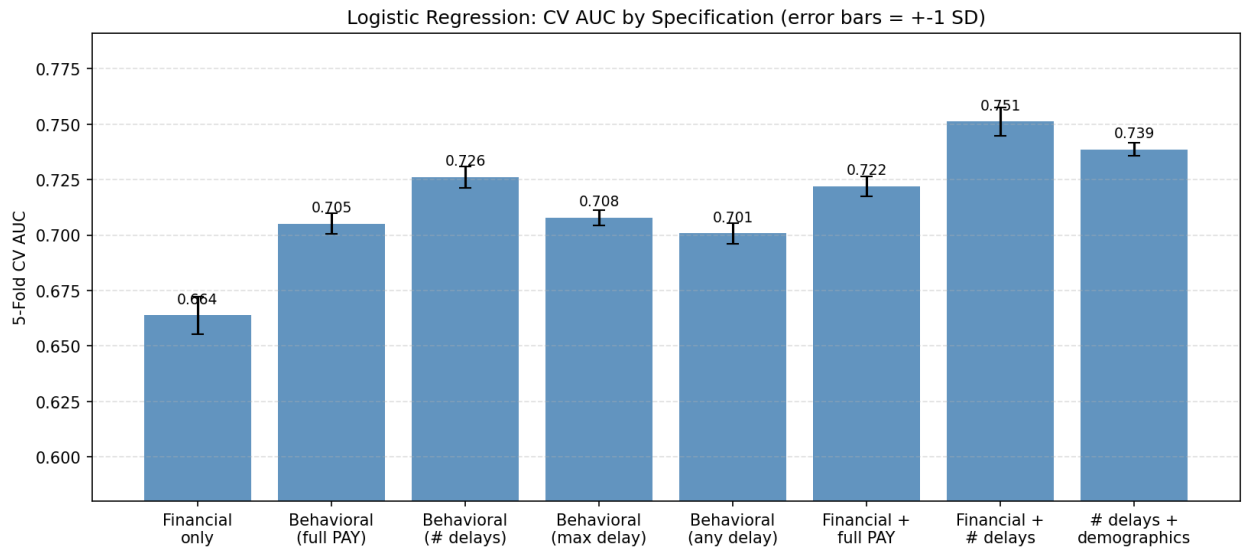


Figure 4: CV AUC by covariate specification (± 1 SD across folds).

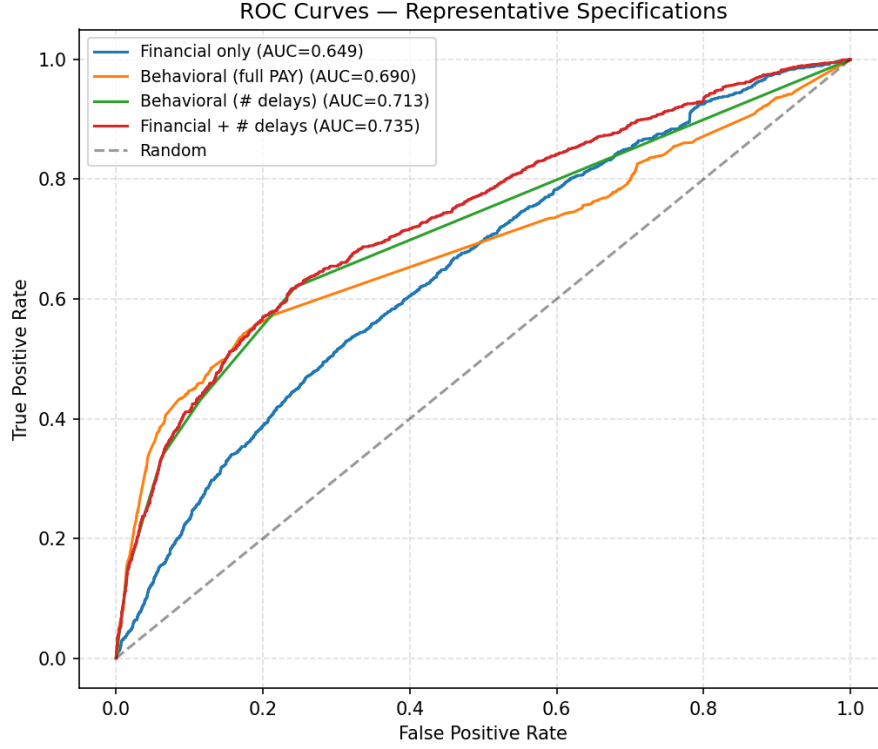


Figure 5: ROC curves for representative logistic specifications.

9 Regularization Paths

Figures 6 and 7 show coefficient paths across regularization strengths, with variable groups color-coded: PAY behavioral (green), BILL_AMT (orange), PAY_AMT (blue), LIMIT_BAL/summaries (gray dashed).

For LASSO, ANY_DELAY, NUM_DELAY_MONTHS, and PAY_0 enter earliest, carrying signal that survives strong penalization. BILL_AMT1 and BILL_AMT5 enter last. At the CV-optimal point the two largest coefficients belong to NUM_DELAY_MONTHS and LIMIT_BAL, the variables in the best parsimonious specification.

For Ridge, BILL_AMT variables carry near-zero coefficients throughout, consistent with their high multicollinearity. Behavioral variables and LIMIT_BAL retain larger, more stable coefficients. Both methods achieve test AUC of 0.740.

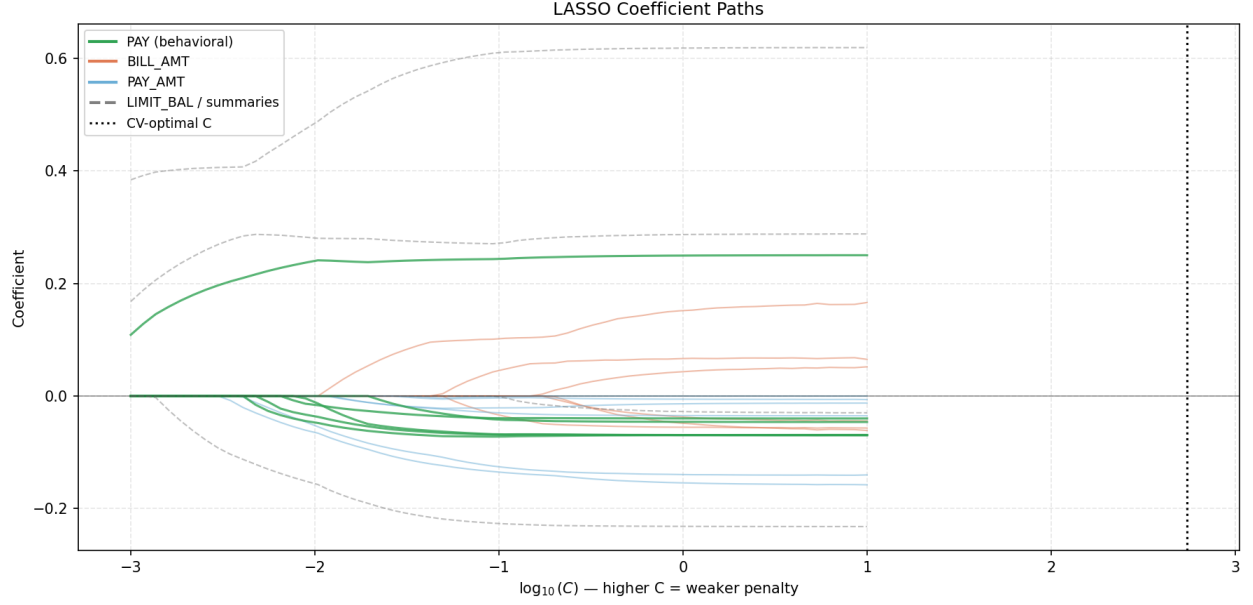


Figure 6: LASSO coefficient paths. Green: PAY behavioral; orange: BILL_AMT; blue: PAY_AMT; gray dashed: LIMIT_BAL/summaries. Dotted line: CV-optimal penalty.

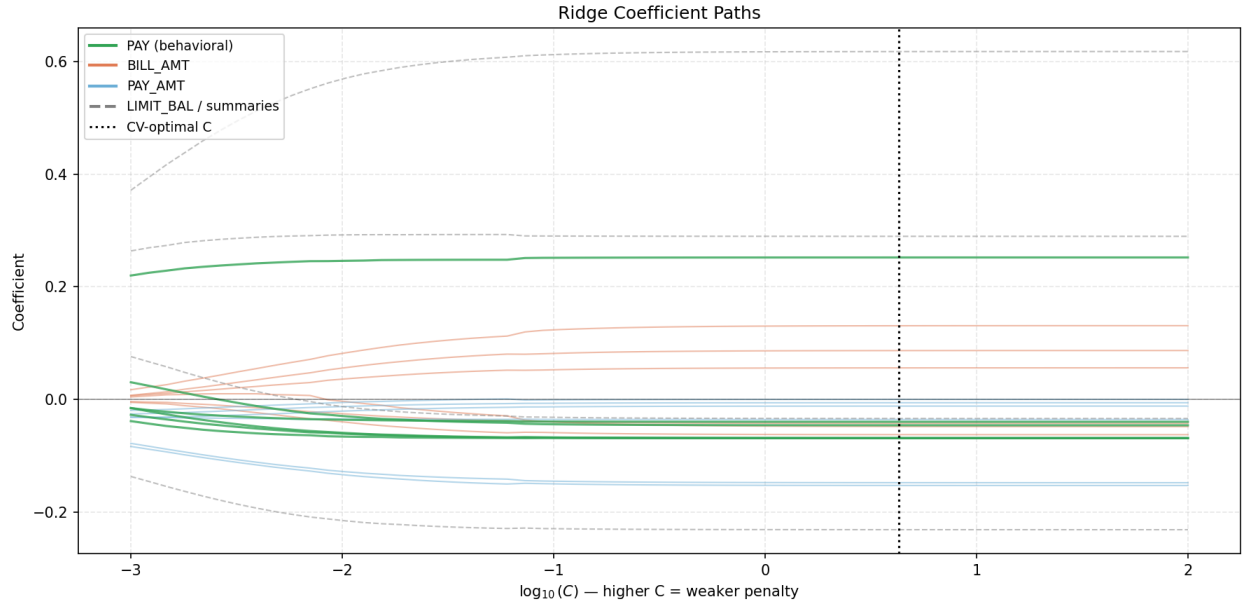


Figure 7: Ridge coefficient paths. BILL_AMT (orange) near zero throughout; behavioral variables (green) and LIMIT_BAL (gray dashed) retain larger magnitudes.

10 Threshold Optimization

Figure 8 shows cost curves and recall/precision tradeoffs for four cost ratios. Under symmetric costs (1:1) the minimum is broad, spanning thresholds from 0.20 to 0.50, so the decision is robust to imprecise calibration. Under asymmetric ratios (5:1, 10:1) the minima sharpen considerably,

meaning small deviations from the optimal threshold carry material cost. Institutions with high loss given default should invest in precise threshold calibration rather than defaulting to 0.5.

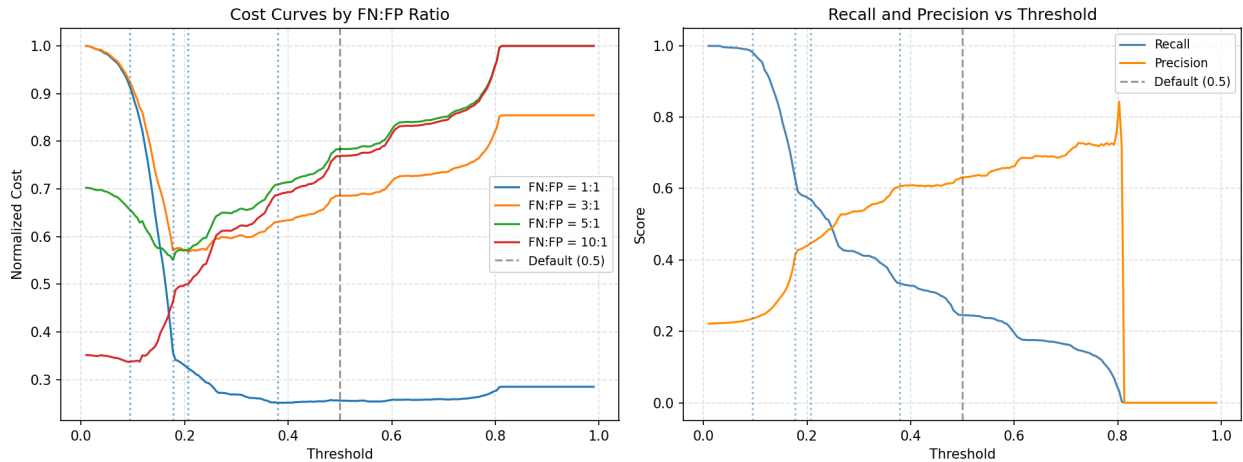


Figure 8: Left: normalized cost curves by cost ratio. Right: recall and precision vs. threshold.

11 Regime Heterogeneity

We embed segment membership directly into the model rather than analyzing regimes separately. A delinquency indicator D_i is added with full interaction terms, producing an unrestricted model where each predictor's contribution varies by regime. The restricted model includes D_i as a main effect only. A likelihood ratio test yields $\Lambda = 63.04$, $df = 13$, $p < 0.001$, rejecting the restricted model.

The predictive improvement is modest: test AUC of 0.737 (restricted) and 0.738 (unrestricted) versus 0.735 for the best unpenalized specification. Two interactions are statistically significant. For `BILL_AMT2` ($p = 0.002$), the coefficient is near zero among non-delinquent borrowers (-0.052) but rises to $+0.724$ among delinquent ones, where a high outstanding balance signals elevated charge-off risk. For `BILL_AMT3` ($p = 0.004$), the sign reverses: $+0.269$ for non-delinquent versus -0.398 for delinquent borrowers, likely reflecting that a prior high balance has since been resolved. The `LIMIT_BAL` interaction is insignificant ($p = 0.891$), confirming credit limit operates similarly across regimes.

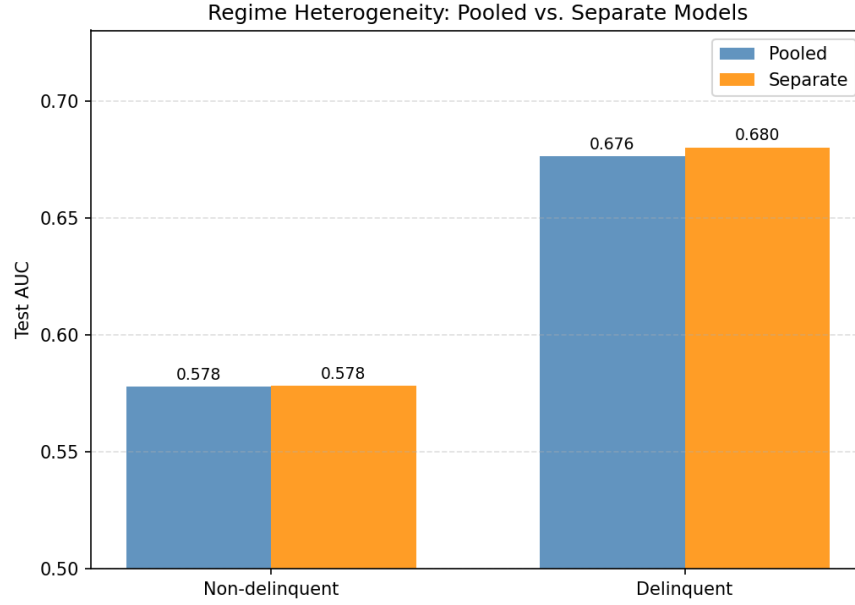


Figure 9: Pooled vs. separate model AUC by delinquency regime.

12 XGBoost Benchmark

The 0.040 AUC gap between XGBoost and the best logistic specification is concentrated at $\text{FPR} \in [0.05, 0.25]$; at higher false positive rates the curves run nearly parallel. For most retail lending use cases operating at moderate recall targets, the logistic model is a credible substitute.

XGBoost feature importances corroborate LASSO: behavioral summaries dominate at 0.572 total importance ($\text{MAX_DELAY} = 0.279$, $\text{ANY_DELAY} = 0.197$, $\text{NUM_DELAY_MONTHS} = 0.096$), while BILL_AMT contributes only 0.061. Both models are comparably well calibrated, meaning the threshold analysis applies equally to both.

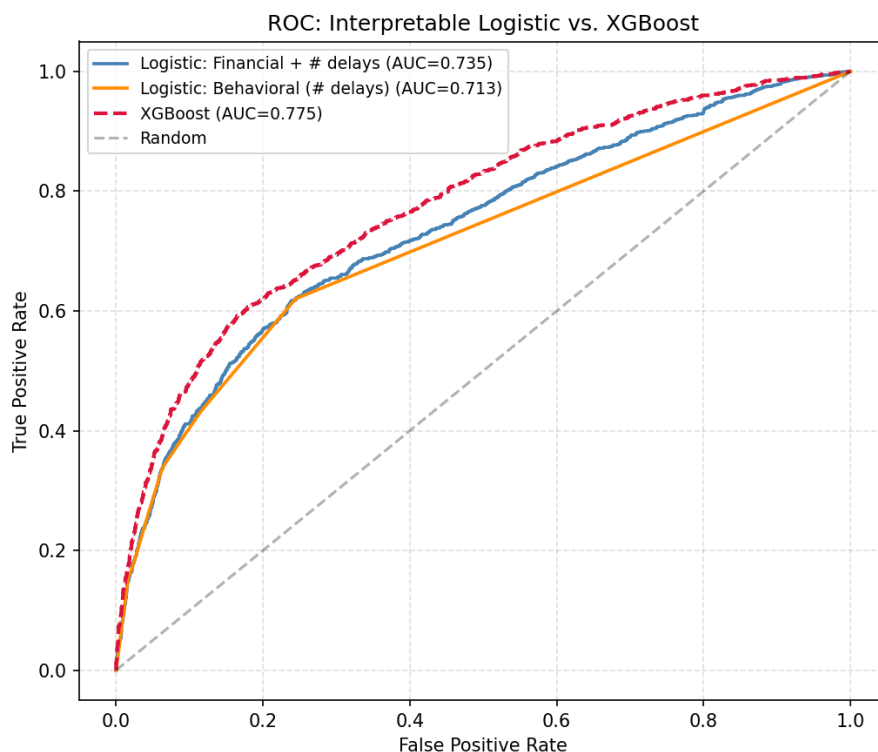


Figure 10: ROC curves: interpretable logistic vs. XGBoost.

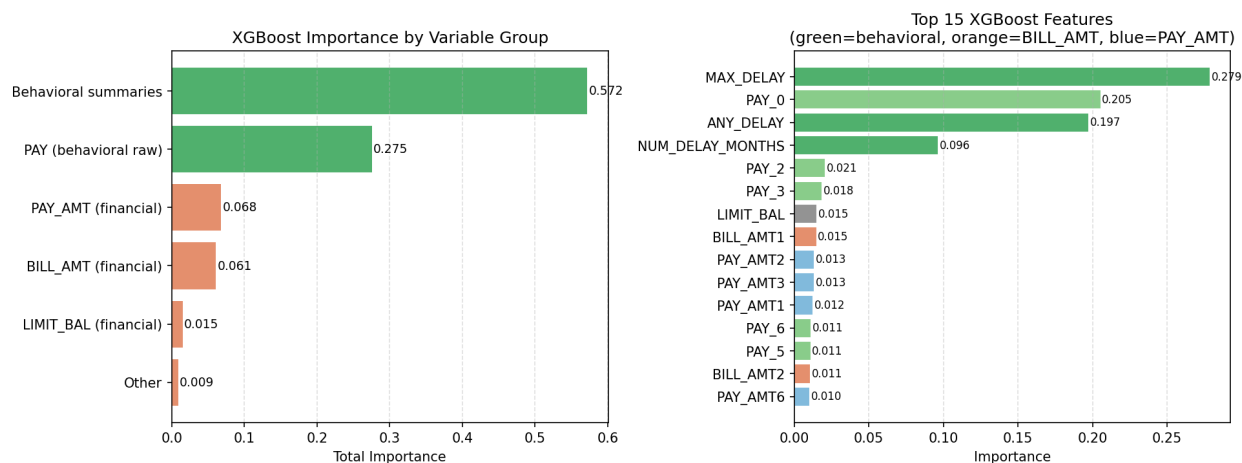


Figure 11: XGBoost feature importance by group (left) and top 15 features (right).

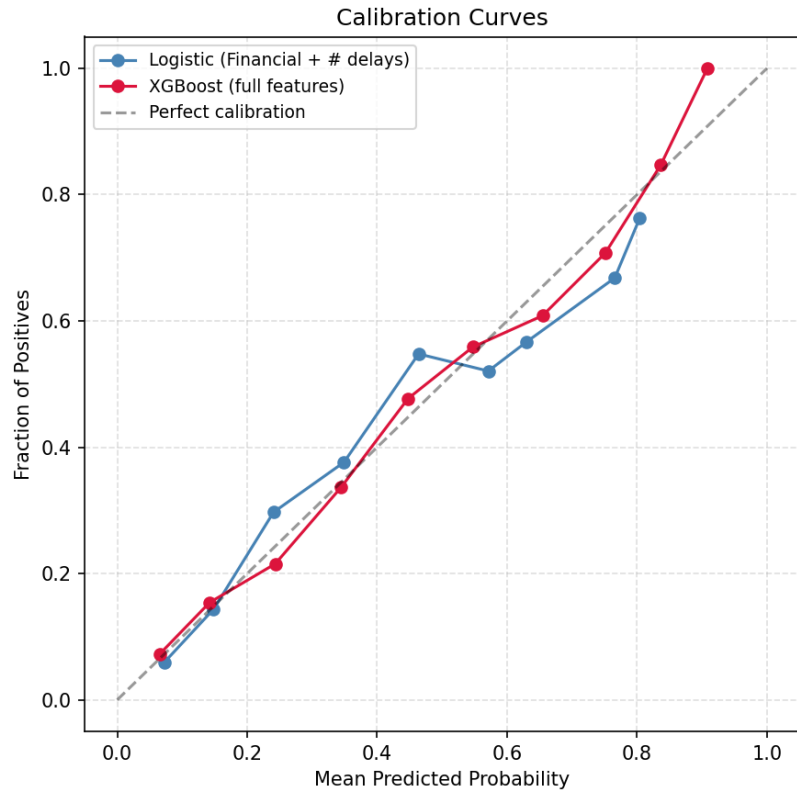


Figure 12: Calibration curves for logistic regression and XGBoost.